

Here is the completely rewritten blueprint, pivoting the architecture strictly to **LiteRT (Mobile On-Device AI)** and structuring the narrative around three powerful problem-to-solution pillars.

This sets you up perfectly for the **\$10,000 LiteRT Special Prize** and the **Digital Equity Track**.

Bina.ai: The Edge-Native App Store for Localized AI

The Concept: A "Shopify for Localized AI Agents" built specifically for the mobile-first, low-resource populations of Southeast Asia.

Instead of forcing rural users to learn "prompt engineering" on expensive cloud platforms, Bina.ai allows them to download specialized, expert-crafted AI agents (like a Midwife Advisor or a Crop Disease Expert) directly to their smartphones. The entire system runs offline via LiteRT.

The 3 Pillars of Bina.ai

Pillar 1: The Accessibility & Cost Barrier

- **The Problem:** High-quality AI is locked behind paywalls, subscription fees, and massive cloud computing costs. Furthermore, marginalized communities in SEA often rely on cheap, low-end Android smartphones and suffer from spotty, expensive internet data.
- **The Solution (LiteRT): Zero-Cost, Offline Edge AI.**
 - We completely bypass cloud API token costs. When a user downloads Bina.ai, a heavily compressed, quantized **Gemma 4 LiteRT model** (approx. 1.5GB) is downloaded once to their phone.
 - From that moment on, all AI generation happens locally on their device's CPU/NPU. It requires zero internet connection to chat with the agent, ensuring complete privacy and absolute accessibility for the poorest demographics.

Pillar 2: The Credibility & Knowledge Gap

- **The Problem:** "Prompt engineering" is difficult. If you give a generic AI to a rural farmer, they won't know how to ask the right questions, and the AI might hallucinate dangerous agricultural or medical advice (the "Garbage In, Garbage Out" problem).
- **The Solution: The "Skill Hub" & Expert Recipes.**
 - We bridge the gap between human professionals and end-users. NGOs, university professors, and certified doctors act as "Creators." They author highly credible **"Agent Recipes"** (lightweight JSON configuration files containing precise system prompts, guardrails, and UI layouts).

- The end-user simply opens the app, browses the Hub, and taps **"Install"** on the *Bidan-Bot (Midwife Advisor)*. The app instantly wraps the expert's credible logic around the local Gemma model.

Pillar 3: The Linguistic Divide

- **The Problem:** Most high-tier AI systems default to English or formal Malay/Indonesian, alienating millions of people who communicate strictly in local dialects (e.g., Iban, Taglish, Kelantanese).
- **The Solution: The Auto-Localization Engine.**
 - A medical professor in Singapore writes the "Health Advisor" recipe entirely in formal English.
 - When an Iban speaker in rural Sarawak installs the recipe, Bina.ai intercepts it. Before launching, the local Gemma model is instructed to seamlessly translate the expert's prompt and the app's user interface buttons into the user's native dialect. The user gets Ivy-League expertise delivered in the language of their hometown.

Recommended MVP Architecture (10-Day Sprint)

To build this for the hackathon, you need to transition from web to mobile development:

1. The Tech Stack

- **Frontend / App Shell:** Flutter or React Native. (Allows you to build a clean mobile UI quickly).
- **The AI Engine:** Google's **MediaPipe LLM Inference API** or the official **LiteRT** libraries for Android. This handles loading the `.flite` or `.task` Gemma model onto the phone.
- **The Database (Skill Hub):** Supabase or Firebase. Because you are only storing tiny text-based JSON "Recipes" (and not the AI model itself), your cloud hosting costs will be literally pennies.

2. The "Recipe" Schema (The One-Tap Install)

The expert writes a JSON file. When the user taps "Install", your Flutter/React Native app reads this file and dynamically generates the UI buttons and system prompt on the screen.

JSON

```
{
```

```
"agent_id": "agri_advisor_01",  
"name": "Pakar Tani (Agri-Expert)",  
"author": "Universiti Putra Malaysia (Verified)",  
"system_prompt": "You are a certified agricultural advisor. Diagnose crop issues based on user  
input. Only recommend locally available Southeast Asian fertilizers. Speak in simple, highly  
accessible language.",  
"ui_buttons": ["Semak Cuaca", "Penyakit Daun", "Harga Baja Semasa"],  
"theme_color": "#2E7D32"  
}
```



The Pitch Narrative for the Judges

*"In Southeast Asia, the AI revolution is leaving the most vulnerable behind. Current tools require fast internet, expensive subscriptions, and the ability to speak English and write complex prompts.

We built **Bina.ai** to democratize not just access to AI, but the creation of it. Using LiteRT, we placed Gemma 4 directly onto the user's mobile device, completely eliminating cloud costs and enabling 100% offline access. We then introduced the Skill Hub—a marketplace where certified professionals upload lightweight 'Recipes'.

A rural user doesn't need to know how to prompt an AI. They just tap 'Install' on an agricultural app authored by a university expert. Bina.ai dynamically generates a native mobile interface and auto-translates the expert's English prompt into the user's local dialect. We are transforming generic LLMs into credible, localized, offline lifelines."*